Foundations of Coding Lab 4 – Statistics Calculator

In this lab you will implement a statistics calculator that is capable of four operations:

- Get Minimum
- Get Maximum
- Get Median
- Get Mean
- Get stdDev

You will do this by writing and implementing four functions, the functions above plus additional helper functions that aid in helping the program run:

- Get List
- Print Menu

In this lab, you will be provided with a template file with all of the function prototypes required to complete the lab.

Below is the expected output of each function:

Get List

```
PS C:\Users\eSports\Documents\275> & C:/Users/eSports/AppData/Local/Programs/Python/Python312/python.exe c:/Users/eSports/Documents/275/lab4.py
Welcome to the List Statistics Calculator
Enter a list of integers or 'q' to end the list!
Enter an integer: 1
Enter an integer: 2
Enter an integer: 3
Enter an integer: 4
Enter an integer: 5
Enter an integer: q
Please choose the statistic you would like to calculate!
```

Print List

```
Please choose the statistic you would like to calculate!
1. Min
2. Max
3. Mean
4. Median
5. Clear List
```

Get Min

```
Please enter your choice, or 0 to exit: 1
The minimum value is: 1
```

Get Max

```
Please enter your choice, or 0 to exit: 2
The maximum value is: 5
```

Get Mean

```
Please enter your choice, or 0 to exit: 3
The mean value is : 3.0
```

Get Median

```
Please enter your choice, or 0 to exit: 4
The median value is : 3
```

Get StdDev

```
the Standard Deviation is: 28.86607004772212
```

Test Cases

```
mylist =
[17,64,16,71,13,92,97,98,31,60,83,2,84,68,62,22,24,76,95,55,6,59,10,29,35,85,26,4
1,53,12,61,20,77,89,5,94,74,25,81,70,82,73,18,15,46,69,3,87,66,23,80,42,21,7,65,9
0,36,96,52,79,32,39,1,33,40,50,14,48,45,51,27,34,30,100,88,9,63,86,19,78,54,47,57
,49,67,91,56,44,58,72,43,38,11,37,75,4,28,99,8,93]
```

Use the list above as the argument for your function to test your standard deviation function